

Requirements Specification Table

Airport Lost Luggage Claim & Tracking Portal (Problem Statement #30)

PES University - Dept. of CSE | SE Lab 1

Target System: Airline Baggage Recovery Platform
Target Stakeholders / Actors: Passenger, Baggage Service Agent

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	Cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals.	High	Pass: Match identified & recovery alert generated. Fail: Incorrect bag tag format accepted.	Enables automated matching of misplaced luggage across transit hubs.
FR-002	Functional	Allow passengers to submit a digital lost baggage claim using Flight Booking Reference (PNR) and bag tag number.	High	Pass: Claim ID generated with status 'Submitted'. Fail: Claim submitted without mandatory bag tag number.	Eliminates manual queueing at physical service desks during baggage loss.
FR-003	Functional	Display a real-time scan timeline showing multi-leg baggage transit locations and handling statuses.	Medium	Pass: Multi-leg scans display correct timestamp & airport code. Fail: Timeline omits intermediate transit scan.	Provides transparent tracking to passengers and reduces support inquiry volume.
FR-004	Functional	Allow Baggage Service Agents to manually override and update baggage scan status and transit locations.	High	Pass: Status updates reflect in portal within 5 seconds. Fail: Unauthorized agent overrides status log.	Empowers ground staff to update routing manually during offline exceptions.
FR-005	Functional	Automatically initiate compensation calculation workflows for baggage delayed beyond 24 hours.	Medium	Pass: Compensation claim opened automatically at 24h mark. Fail: Delayed bag requires manual system trigger.	Ensures compliance with automated passenger compensation protocols.
NFR-001	Performance & Security	The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms under peak load.	High	Pass: Benchmarking tests confirm target latency (<200 ms). Fail: Latency exceeds target standard.	Ensures high system responsiveness during peak travel disruption events.
NFR-002	Security & Privacy	All passenger claim records and Personally Identifiable Information (PII) must be encrypted at rest (AES-256) and in transit (TLS 1.3).	High	Pass: Security audit confirms active end-to-end encryption. Fail: Unencrypted PII detected in network log.	Protects sensitive passenger personal data against breaches and regulatory non-compliance.

UML Use-Case Modelling & Structure

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Target System: Airport Lost Luggage Claim & Tracking Portal

Primary Actors: Passenger, Baggage Service Agent

1. System Actors

- **Passenger (Primary Actor):** End-user filing baggage claims and checking tracking timelines.
- **Baggage Service Agent (Secondary Actor):** Airport staff updating scan logs and processing exceptions manually.

2. Use Cases Summary

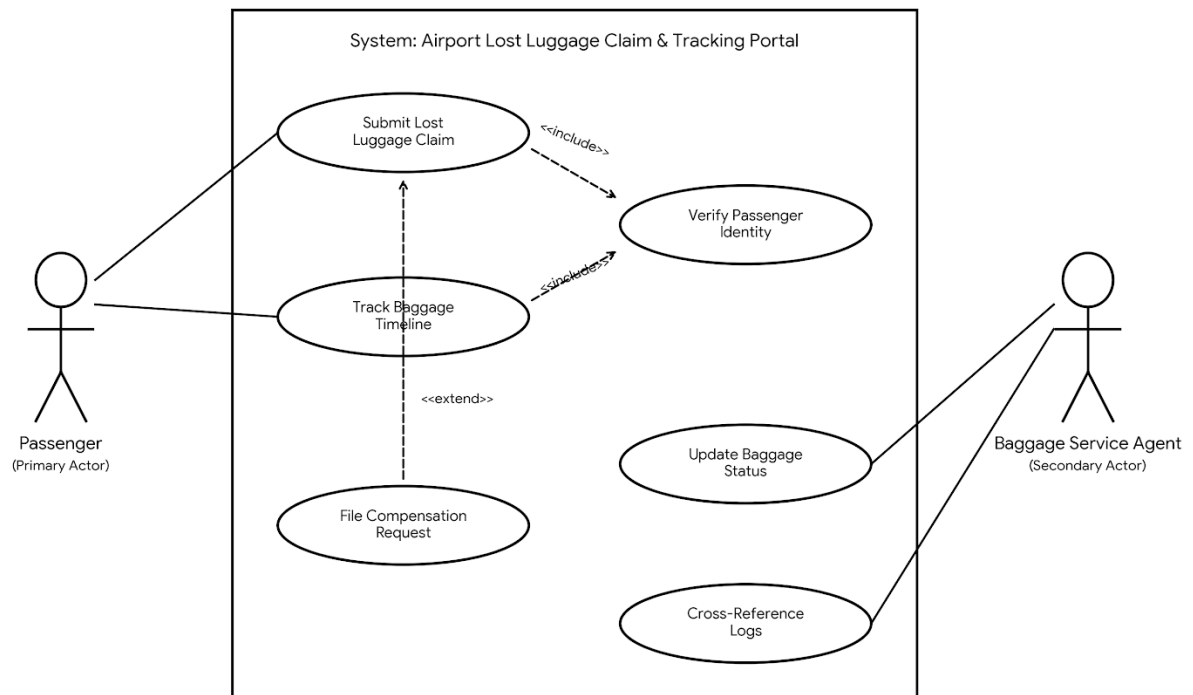
- **UC-01: Submit Lost Luggage Claim** (Passenger)
- **UC-02: Track Baggage Timeline** (Passenger)
- **UC-03: Verify Passenger Identity** (System Security)
- **UC-04: File Compensation Request** (Passenger)
- **UC-05: Update Baggage Location & Status** (Agent)
- **UC-06: Cross-Reference Unrouted Logs** (Agent)

3. Use-Case Diagram Relationships

- **«include» Relationship:**
UC-01 (Submit Claim) and UC-02 (Track Timeline) **«include»** UC-03 (Verify Passenger Identity).
Rationale: Every claim submission or tracking lookup strictly requires verifying passenger credentials prior to processing.
- **«extend» Relationship:**
UC-04 (File Compensation Request) **«extend»** UC-01 (Submit Claim).
Rationale: Compensation filing is an optional extension triggered only when baggage delay exceeds the 24-hour threshold.

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UML Use-Case Diagram | Problem Statement #30 (PES University SE Lab 1)



Use-Case Flow Specification

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Use Case ID & Name: UC-01 Submit Lost Luggage Claim

Primary Actor: Passenger

Secondary Actor: Baggage Service Agent

Target System: Airport Lost Luggage Claim & Tracking Portal

Preconditions:

1. Passenger possesses a valid Flight Booking Reference (PNR) and Bag Tag number.
2. Portal is active and connected to global transit scan databases.

Postconditions:

1. A new Baggage Claim record is initialized with status 'Submitted' and assigned a Claim Reference ID.
2. Automated background search (FR-001) is triggered across connected airport terminals.

Main Success Scenario:

1. Passenger navigates to the portal and selects "File Lost Baggage Claim".
2. System prompts passenger to enter PNR, Bag Tag Number, and Contact Details.
3. Passenger inputs required details and submits the form.
4. System authenticates passenger credentials («include» UC-03: Verify Passenger Identity).
5. System validates bag tag format and verifies against flight manifest.
6. System generates Claim Reference ID, sends confirmation notification, and starts automated cross-referencing.

Alternate Flow (A1: Invalid Bag Tag Format):

1. At step 5, system identifies invalid bag tag format.
2. System displays error message: "Invalid Bag Tag format. Please enter a valid 10-digit numeric tag."
3. Passenger re-enters correct bag tag number and resumes from step 4.